

Project Apex Race Report - Race Event

Executive Summary

Wayne Taylor Racing (Car #1, Lamborghini) and World Speed (Car #22, Lamborghini) clearly lead the field. Car #1 demonstrated superior raw pace with an optimal lap of 2:05.207 and a competitive average green flag pace of 2:12.607. Car #22 showcased exceptional consistency with a 0.261 standard deviation in race pace and highly efficient pit stops (1:36.673 average stationary time). These teams effectively managed tire degradation, with Car #1 showing a "FAIR" model quality and low predicted final 5 laps loss (0.834765s). MLT Motorsports (Car #11, Lamborghini), ANSA MOTORSPORTS (Car #4, Lamborghini), and TR3 Racing (Car #29, Lamborghini) form a competitive mid-field. Car #11 stands out with strong pace consistency (1.449 stdev) and efficient pit stops (1:37.853 stationary time). Car #29 also displayed strong green flag pace (2:13.381) and reasonable tire degradation (1.1434279999999999s end-of-stint deg rate). The key challenge for this tier lies in optimizing driver performance consistency across the lineup and refining tire management strategies to unlock full potential. Teams like RAFA Racing (Car #99, Lamborghini) and Kaizen Autosport (Car #44, Lamborghini) significantly lagged the field. Car #99 exhibited the slowest average green flag pace (2:47.822) and suffered from exceptionally high pit cycle loss (324.5248s), compounded by a very long average pit stationary time (2:57.817). Car #44 also faced substantial pit cycle losses (247.77175s). A pervasive issue across many lagging teams, particularly within the ProAm and Am classes, was the significant performance disparity between drivers in the same car (e.g., Car #46 with a 22.048s delta, Car #2 with 21.351s). The ability to minimize driver performance deltas within multi-driver cars and execute consistently fast pit stops proved to be the most significant strategic differentiators in this race, often outweighing marginal gains from raw pace or tire management for many teams.

Tactical Insights

- {'team': 'Wayne Taylor Racing', 'car_number': '1', 'category': 'Leading Team', 'recommendation': "Enhance overall team performance by narrowing the significant pace gap between primary and secondary drivers, thereby maximizing the car's potential across all stints.", 'justification': "While Car #1's lead driver, Danny Formal, achieved an optimal lap of 2:05.207, the team's second driver, Hampus Eriksson, had a best lap 17.513s slower, as detailed in driver_deltas_by_car. This substantial delta indicates a considerable amount of untapped potential and lost track time during Eriksson's stint. Addressing this disparity through

targeted coaching, simulation work, or driver lineup adjustments would solidify their leading position and reduce vulnerability over longer race distances."}

- {'team': 'MLT Motorsports', 'car_number': '11', 'category': 'Mid-field Team', 'recommendation': 'Capitalize on existing strong pace consistency and pit efficiency by refining tire degradation management and optimizing traffic navigation.', 'justification': 'Car #11 demonstrates commendable race pace consistency (1.449 stdev) and efficient pit stops (1:37.853 average stationary time). However, their tire degradation model quality is rated "POOR" in enhanced_strategy_analysis, suggesting an inconsistent or suboptimal tire management strategy over full stints. Additionally, traffic_management_analysis shows Darius Trinka incurred 2 out-of-class traffic laps, indicating potential for lost time. A deeper analysis into their tire data (deg_coeff_a, deg_coeff_b) and focused driver training on traffic management could convert their strong foundation into consistent front-running pace.'}
- {'team': 'RAFA Racing', 'car_number': '99', 'category': 'Lagging Team', 'recommendation': 'Implement immediate and drastic improvements in pit stop execution and overall race pace consistency to establish a foundational competitive baseline.', 'justification': 'Car #99 exhibits critical weaknesses in fundamental race operations. full_pit_cycle_analysis reveals an average pit cycle loss of 324.5248s, significantly higher than the field average, with an average stationary time of 2:57.817s (over a minute longer than leading teams). Furthermore, their average green flag pace (2:47.822) and race pace consistency (37.927 stdev) are among the worst in the field according to enhanced_strategy_analysis. These issues represent a substantial handicap that must be addressed through rigorous pit crew training, strategy re-evaluation, and driver development before any other performance optimizations can be effective.'}